

AC Circuit Analysis



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Circuit with Resistors

- We know resistors don't change the phase of the voltage or current.
- We can just use these equations to work out the values in a resistive circuit
 - Voltage = Current * Resistance
 - Power = Voltage * Current



$$V = I * R$$

$$120 = I * 100$$

$$I = 1.2A$$

$$P = V * I$$

$$P = 120 * 1.2$$

$$P = 144W$$

Circuits with Inductors

- For circuits with inductors, we have a phase shift
- This means we need to change our equations:
 - Inductive Reactance = 2π * Frequency * Reactance
 - Voltage = Current * Inductive Reactance



$$X_L = 2\pi fL$$
$$X_L = 2\pi * 1000 * 5$$
$$X_L = 31415.92654$$

$$V = IX_L$$
$$120 = I * 31415.92654$$
$$I = 0.003819718634$$

Circuits with Capacitors

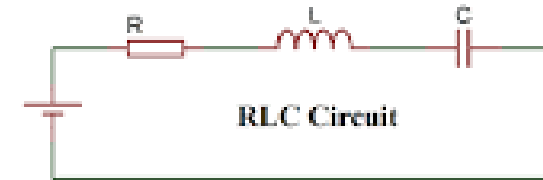
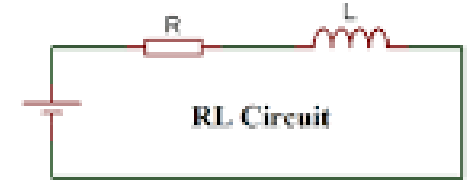
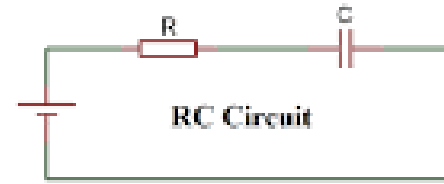
- For circuits with capacitors, we again have a phase shift
- This means we need to use the equations:
 - Capacitive Reactance = $1/2\pi * \text{frequency} * \text{capacitance}$
 - Voltage = Current * Capacitive Reactance



$$X_C = \frac{1}{2\pi f C}$$
$$X_C = \frac{1}{2\pi * 1000 * 5}$$
$$X_C = 3.18309886 * 10^{-5}$$
$$V = I X_C$$
$$120 = I * 3.18309886 * 10^{-5}$$
$$I = 3769911.184$$

RLC, RC and RL Circuits

- RLC Circuits contain all 3 components;
 - A Resistor (R)
 - An Inductor (L)
 - A Capacitor (C)
- RL Circuits contain two of the components;
 - A Resistor (R)
 - An Inductor (L)
- RC Circuits contain two of the components;
 - A Resistor (R)
 - A Capacitor (C)

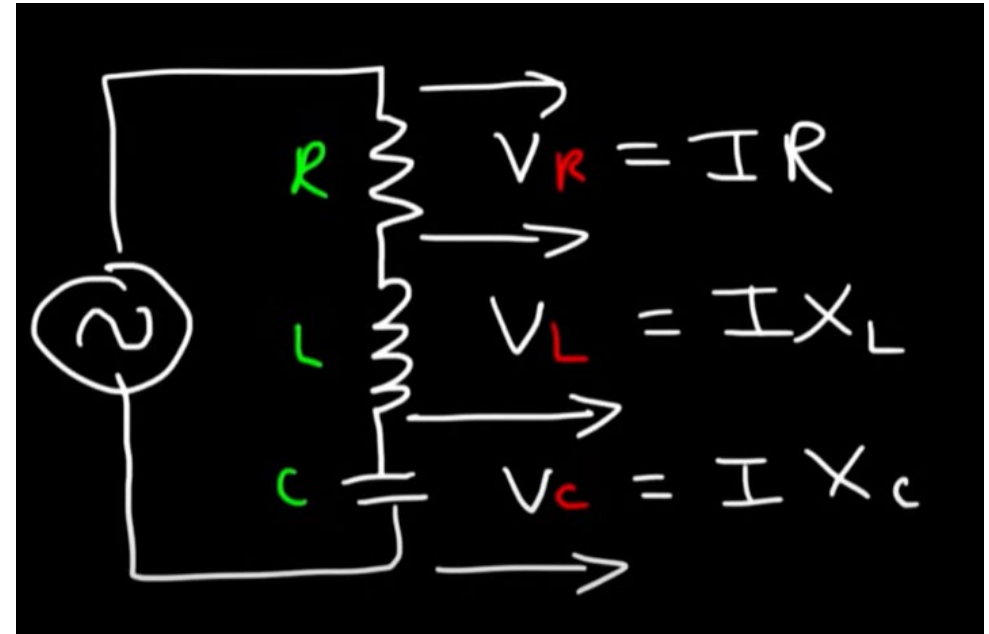


Simple Series RL, RC and RLC Circuits

- For each of the different series circuits you can use impedance to work out voltage and current using the equation:
 - $Z = \sqrt{R^2 + (X_L - X_C)^2}$ where:
 - Z = Impedance
 - R = Resistance
 - X_L = Inductive Reactance
 - X_C = Capacitive Reactance
 - Note when you don't have an inductor or a capacitor you just set them to 0

Series RL, RC and RLC Circuits

- We can then work out Voltage or Current for the whole circuit using the equation
 - $V = IZ$
- However, for each component in the circuit we use their individual equation

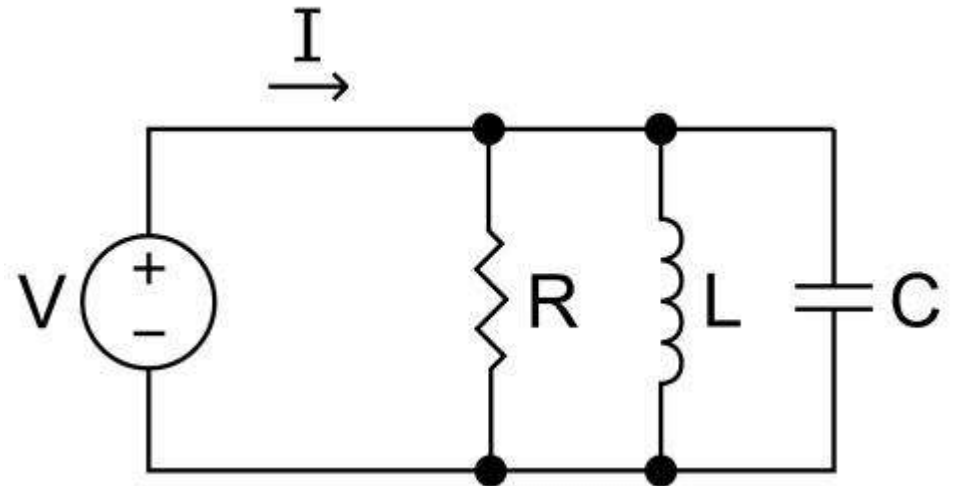


Simple Parallel RLC Circuits

- When we have components in parallel, we work out our impedance differently
- For standard single component circuits:

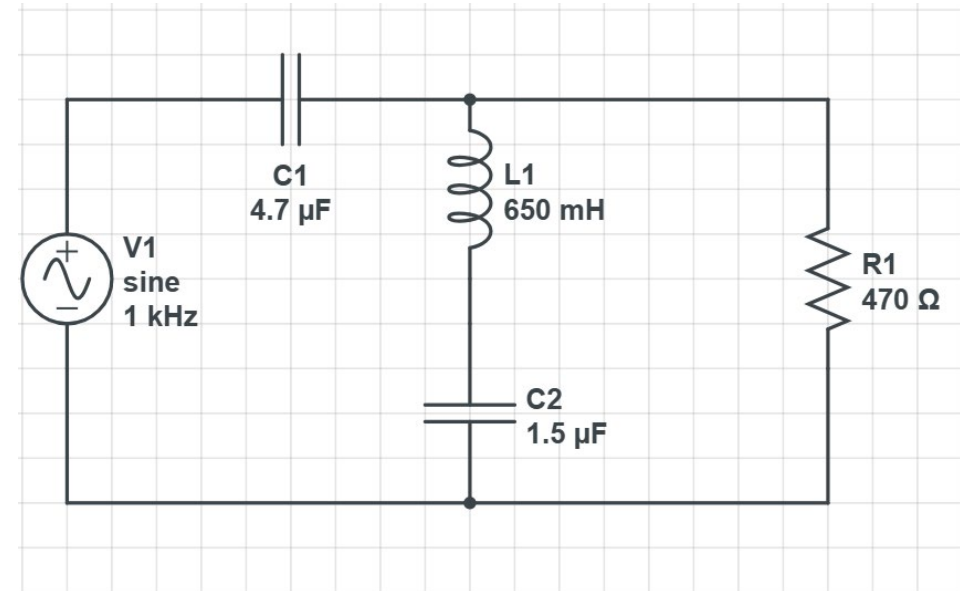
- We use the equation: $Z = \frac{1}{\sqrt{\frac{1}{R^2} + (\frac{1}{X_L} - \frac{1}{X_C})^2}}$

- Remembering we can sub in 0 if we are missing a capacitor or an inductor



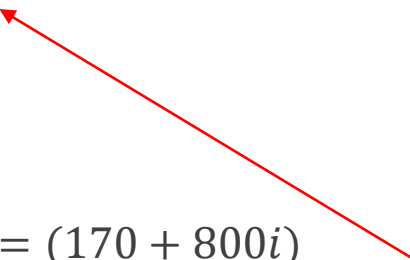
Complex RLC Circuits

- Complex RLC circuits are circuits in which multiple different components make a more advanced circuit rather than just single components in parallel or series
- In general, I'd recommend always working in rectangular when doing complex RLC analysis
- Though you will have to convert to polar to find current and voltage values



Complex Series RLC Circuits

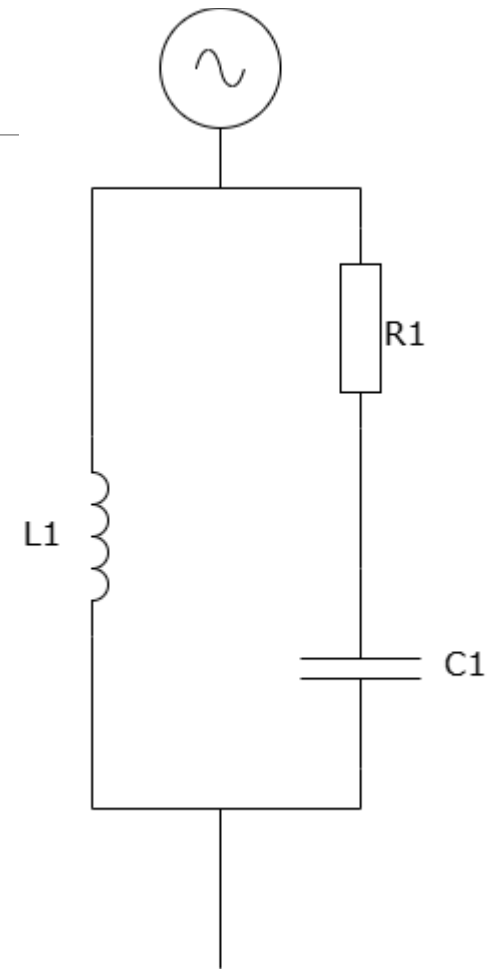
- When we have components in series in a complex circuit, we just add their resistance values together and then their reactance values together
- $(R_1 + i_1) + (R_2 + i_2) = ((R_1 + R_2) + (i_1 + i_2))$
- So for example if we have:
- $(120 + 320i) + (50 + 480i)$
- We get:
- $(120 + 320i) + (50 + 480i) = (170 + 800i)$



Note these are likely a combined components as they both have R and i values

Complex Parallel RLC Circuits

- For more complex RLC circuits we need to use $\frac{1}{\frac{1}{(R_1+i_1)} + \frac{1}{(R_2+i_2)}}$
- For this we need to use admittance $\left(\frac{1}{(R+i)}\right)$
- To work out the admittance we use: $\frac{1}{(R+i)} = \frac{R-i}{R^2+i^2}$ Remember to flip the sign here
- So for instance $(560 + 200i)$ becomes:
- $\frac{1}{(560 + 200i)} = \frac{560 - 200i}{560^2 + 200^2} = \frac{560 - 200i}{353600} = 0.001584 - 0.0005656i$



Complex Parallel RLC Circuits

- So for this example: $(120 + 320i) || (50 + 480i)$

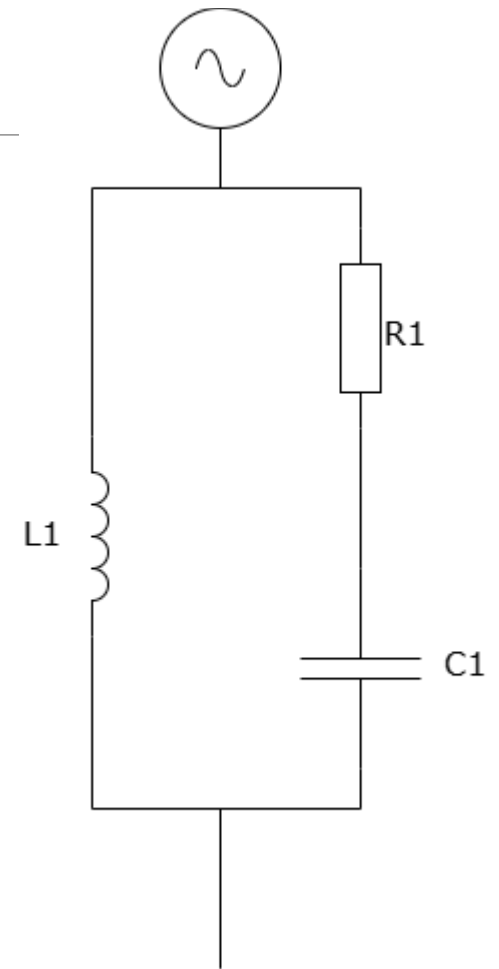
Note the || just means in parallel with

- We do $\frac{1}{\frac{1}{120+320i} + \frac{1}{50+480i}}$

- So first we work out the admittance of each of the parts:

- $\frac{1}{120+320i} = \frac{120-320i}{120^2+320^2} = \frac{120-320i}{116800} = 0.001027 - 0.002740i$

- $\frac{1}{50+480i} = \frac{50-480i}{50^2+480^2} = \frac{50-480i}{232900} = 0.0002147 - 0.002061i$

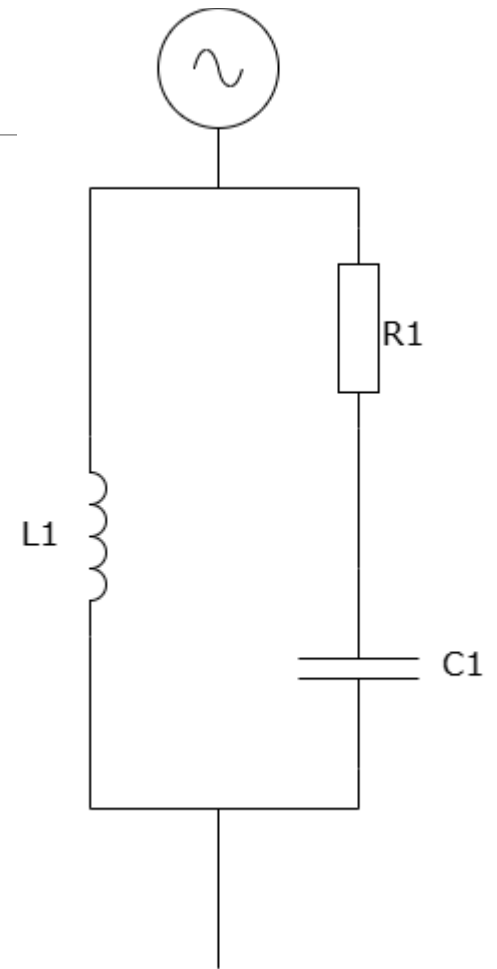


Complex Parallel RLC Circuits

- Now we add the two admittances together
- $(0.001027 - 0.002740i) + (0.0002147 - 0.002061i)$

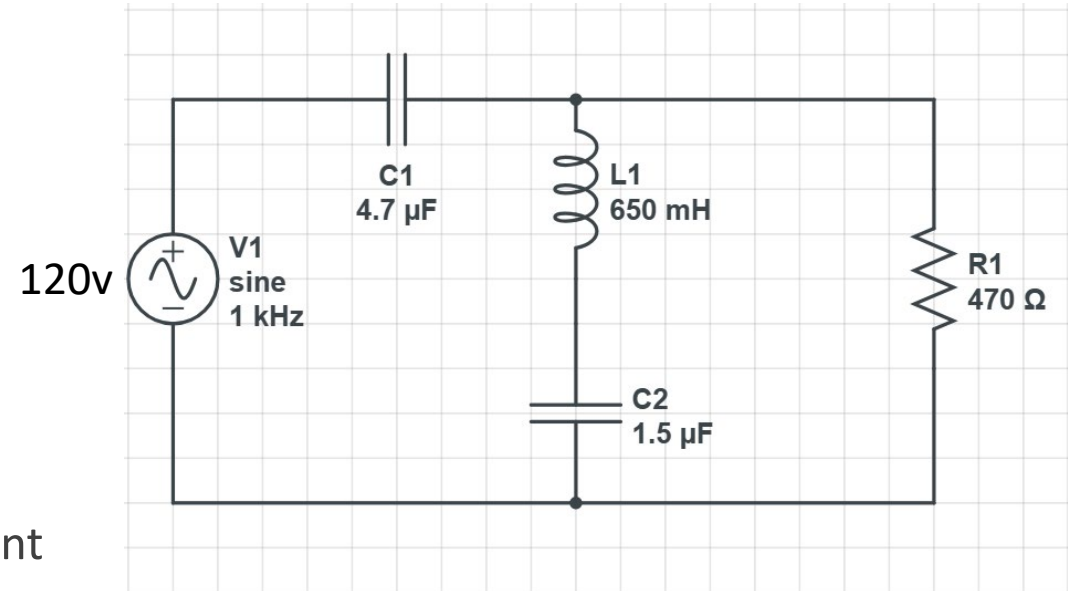
$$(0.001242 - 0.004801i)$$

- Then we convert back to impedance
- $\frac{1}{(0.001242 - 0.004801i)} = \frac{0.001242 + 0.004801i}{0.001242^2 + 0.004801^2} = \frac{0.001242 + 0.004801i}{0.00002459}$
- $(50.5039 + 195.2248i)$



Example Question

- In this question we want to find the voltage and current through every component
- The best strategy to take are to:
 1. Work out the reactance of each component
 2. Combine components to make equivalent circuits
 3. Continue combining components until the circuit is one equivalent component
 4. Expand circuit back out calculating V and I for the components or groups as we go



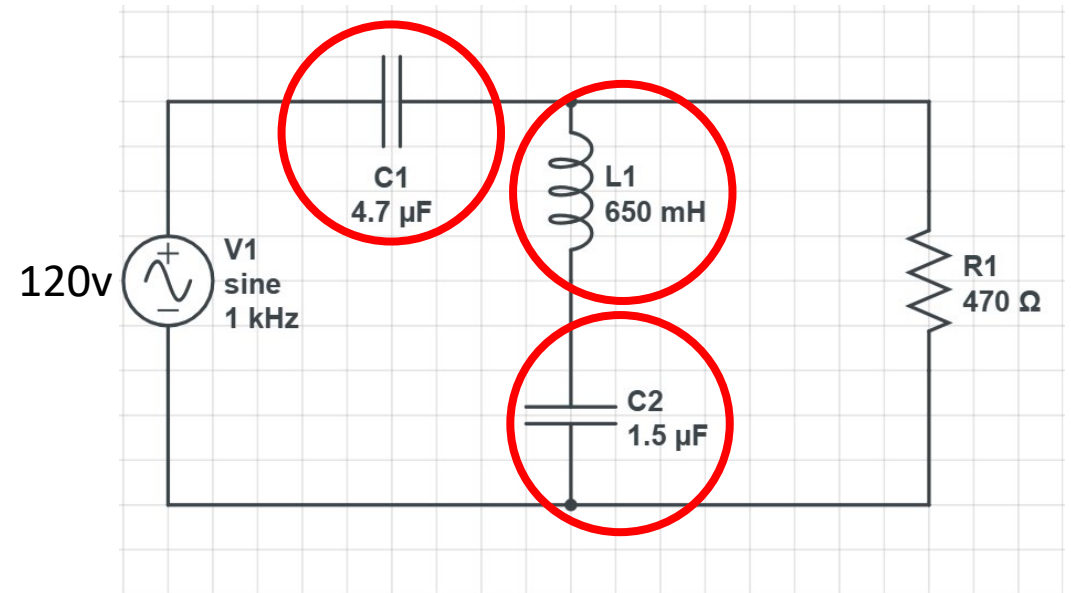
Example Question

- Let's start by calculating the reactance for each component

- $$X_{C1} = \frac{1}{2\pi f C} = \frac{1}{2\pi * 1000 * 0.0000047} = 33.86275385$$

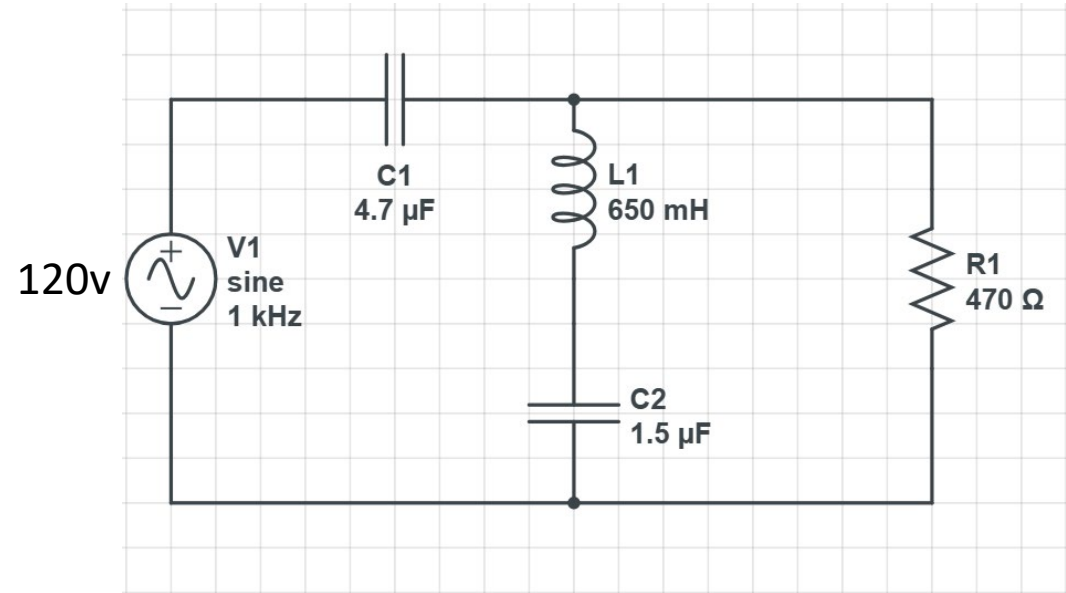
- $$X_{C2} = \frac{1}{2\pi f C} = \frac{1}{2\pi * 1000 * 0.0000015} = 106.1032954$$

- $$X_{L1} = 2\pi f L = 2\pi * 1000 * 0.65 = 4084.07045$$



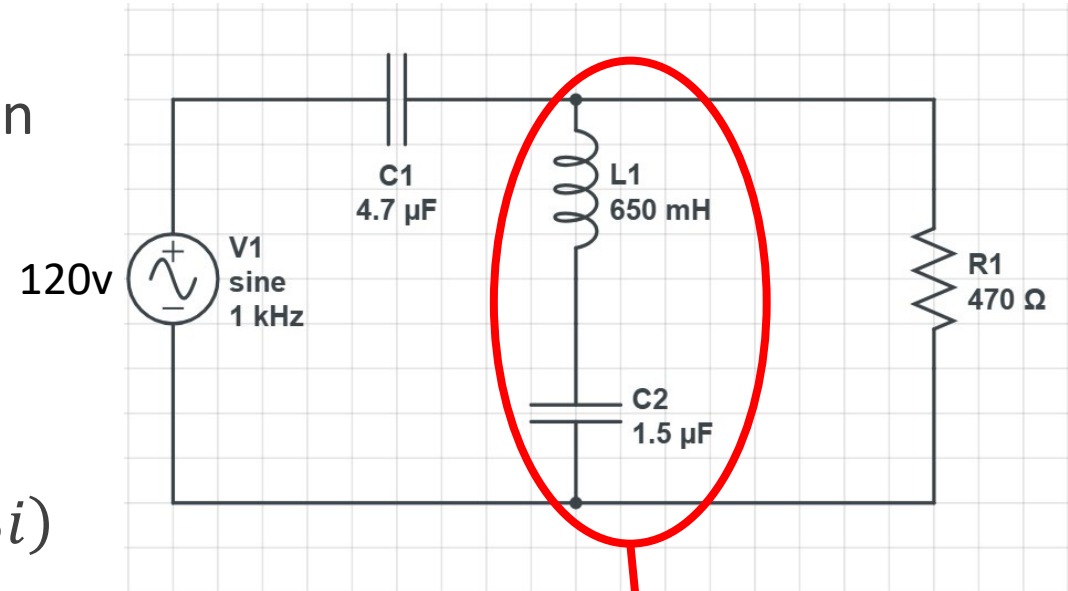
Example Question

- Next let's write out our reactance's as impedances
 - $Z_{C1} = 33.863 \angle -90^\circ$
 - $Z_{C2} = 106.103 \angle -90^\circ$
 - $Z_{L1} = 4084.070 \angle 90^\circ$
 - $Z_{R1} = 470 \angle 0^\circ$

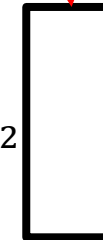


Example Question

- Next let's combine L1 and C2 into an equivalent component
- $Z_{L1} + Z_{C2}$
- $(0 + 4084.070i) + (0 - 106.103i)$
- $(0 + 3,977.967i) = 3977.967\angle 90$



$Z_{L1} + Z_{C2}$



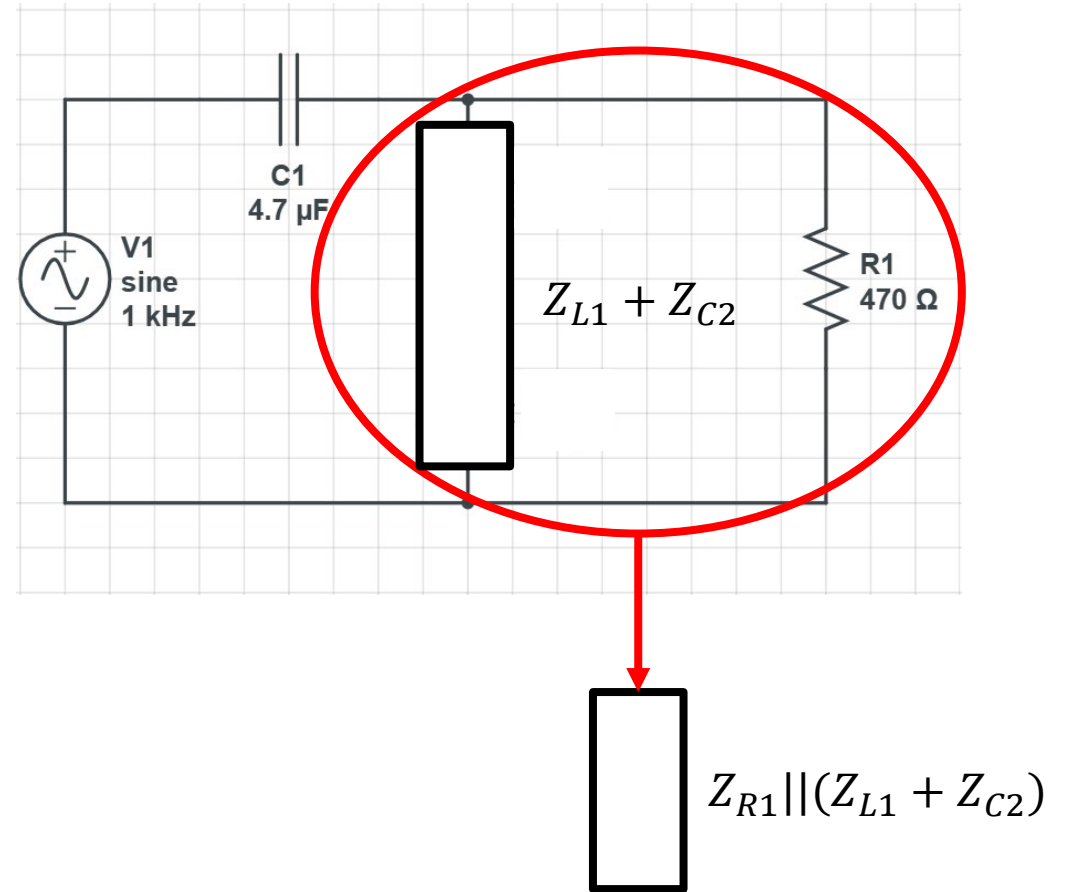
Example Question

- Next let's combine R1 with our equivalent component

- $Z_{R1} || (Z_{L1} + Z_{C2})$

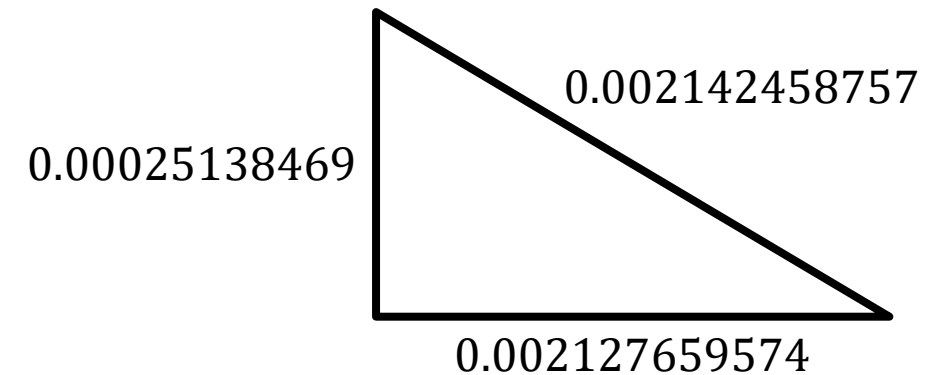
- $$\frac{1}{\left(\frac{1}{470\angle 0} + \frac{1}{3977.967\angle 90}\right)}$$

- $$\frac{1}{(0.002127659574\angle 0 + 0.00025138469\angle 90)}$$



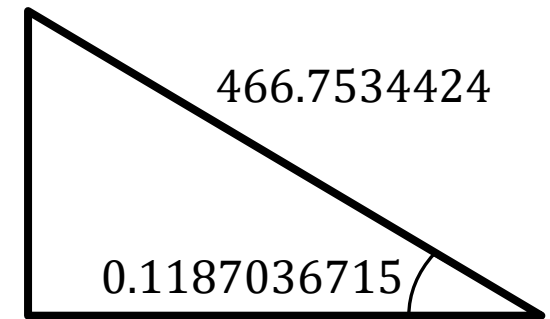
Example Question

- $\frac{1}{(0.002127659574 \angle 0 + 0.00025138469 \angle 90)}$
- $\sqrt{0.002127659574^2 + 0.00025138469^2}$
- $\frac{1}{0.002142458757} = 466.7534424$
- $\tan^{-1}\left(\frac{0.00025138469}{0.002127659574}\right) = 6.738303605$
- $466.7534424 \angle 6.738303605$



Example Question

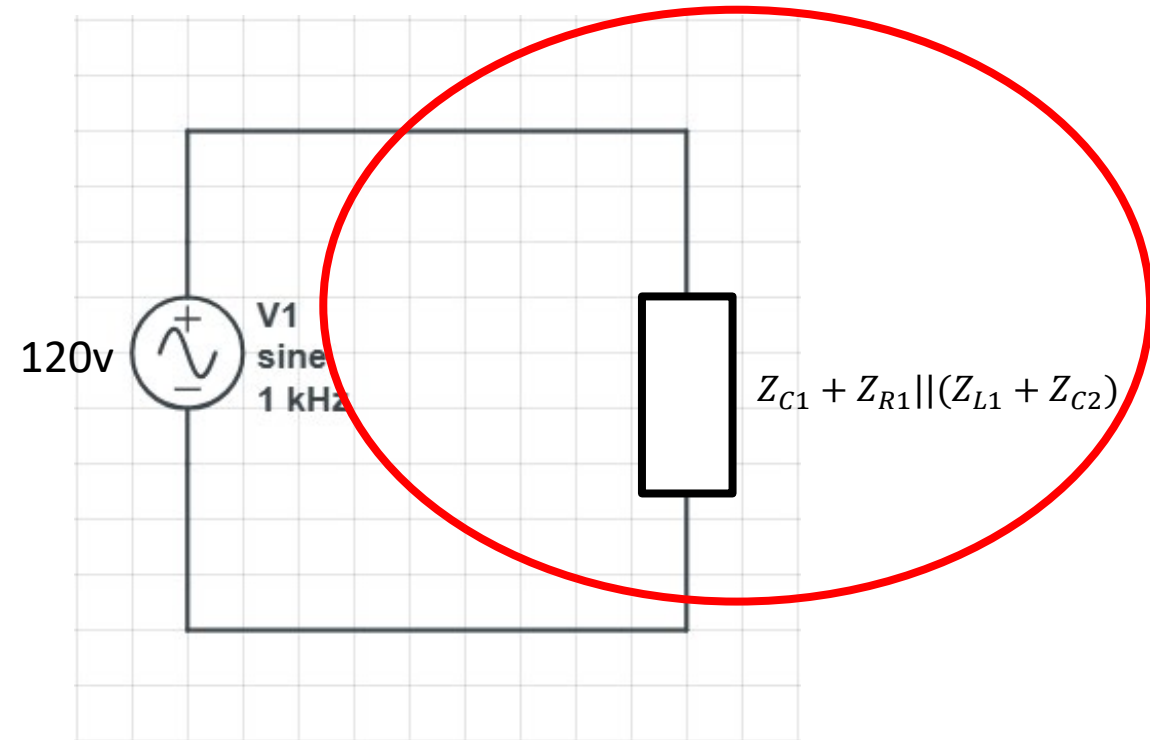
- $466.7534424 \angle 6.738303605$
- Let's convert this back to rectangular form
- $466.7534424 * \sin(6.738303605) = 54.76636086$
- $466.7534424 * \cos(6.738303605) = 463.5293105$
- $(463.5293105 + 54.76636086i)$



$$s_h^o c_h^a t_a^o$$

Example Question

- Let's combine our final two components into an equivalent component
- $Z_T = Z_{C1} + Z_{R1} || (Z_{L1} + Z_{C2})$
- $(0 + 33.86275385i) + (463.4688928 + 55.27532353i)$
- $(463.4688928 + 89.13807738i)$
- $\sqrt{463.4688928^2 + 89.13807738^2} = 471.9629344$
- $\tan\left(\frac{89.13807738}{463.4688928}\right) = 0.1947351008$
- $471.9629344 \angle 0.1947351008$



Example Question

- Let's work out the current going through this combined component

- $V = IZ$

- $I = \frac{V}{Z}$

- $I = \frac{120\angle 0}{471.9629344\angle 0.1947351008} = 0.25425725465$

